

Test Procedure TP-17 - 900 s Controller Demonstration

Revision	C
Date	03 Apr 2026
Status	Released
Test object	PLC-101 sequence behavior

1. Objective

Verify the two-tank controller normal sequence, STOP/START resume behavior and controlled SHUT behavior in a 900 s simulation.

2. Test Setup

Item	Value
Initial T1 level	0.05 m
Initial T2 level	0.05 m
Simulation duration	900 s
Logging interval	1 s
Configuration authority	Approved requirements + DR-02 + CR-004

3. Operator Command Schedule

Time (s)	Command	Expected immediate controller response
20	START	Begin or resume automatic operation; first occurrence begins T1 fill.
220	STOP	Close all three valves and hold sequence context.
280	START	Resume the state interrupted at 220 s.
650	STOP	Close all valves; hold sequence context.
700	SHUT	Enter controlled shutdown, inhibit V1, and command V2 and V3 open until both tanks are at low level.

4. Acceptance Criteria

AC-01 - T1 shall reach the effective high-level setpoint of 0.80 m before the first transfer begins.

AC-02 - All valves shall be closed throughout the STOP interval from 220 s until the START command at 280 s.

AC-03 - The START at 280 s shall resume the interrupted transfer rather than restart filling.

AC-04 - Following completion of the first T2 drain, the controller shall automatically begin another fill cycle after the effective 8 s inter-cycle delay.

AC-05 - All valves shall be closed throughout the STOP interval from 650 s until SHUT at 700 s.

AC-06 - At 700 s V1 shall remain closed and V2/V3 shall be commanded open together for controlled draining.

AC-07 - After shutdown drain completion the controller shall return to IDLE with all valves closed and shall remain there through 900 s because no subsequent START is scheduled.

AC-08 - Normal automatic operation shall never command V1 and V2 simultaneously, nor V2 and V3 simultaneously. The latter combination is permitted only in SHUTDOWN.

5. Expected State Timing Guidance

Exact threshold transition timestamps depend on numerical integration and event ordering. A tolerance of +/-2 s is acceptable for state-boundary comparisons when the effective parameter set is used. The provided CSV is the benchmark trace generated with 1 s logging.